

Spatially Analyzing the Effectiveness of Bluff-protection Structures Installed at Ozaukee County Lake Michigan Bluffs

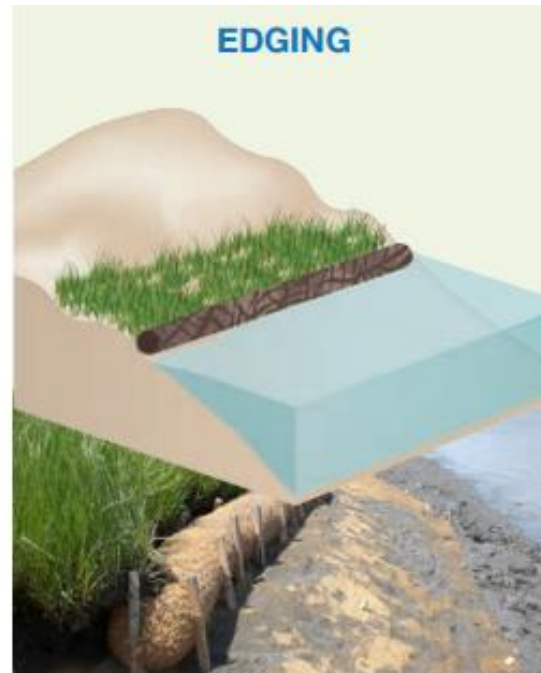
Heriberto Patino Luna - patinoluh02@uww.edu

April 8th, 2024, UMHC annual conference



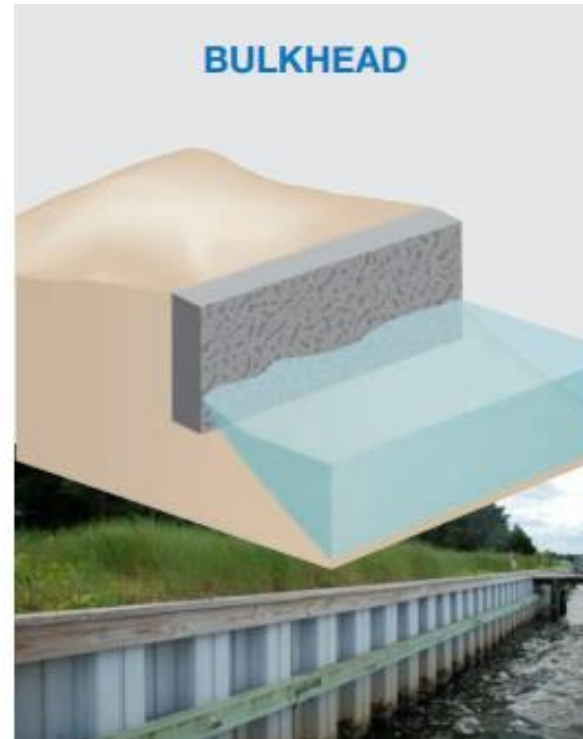
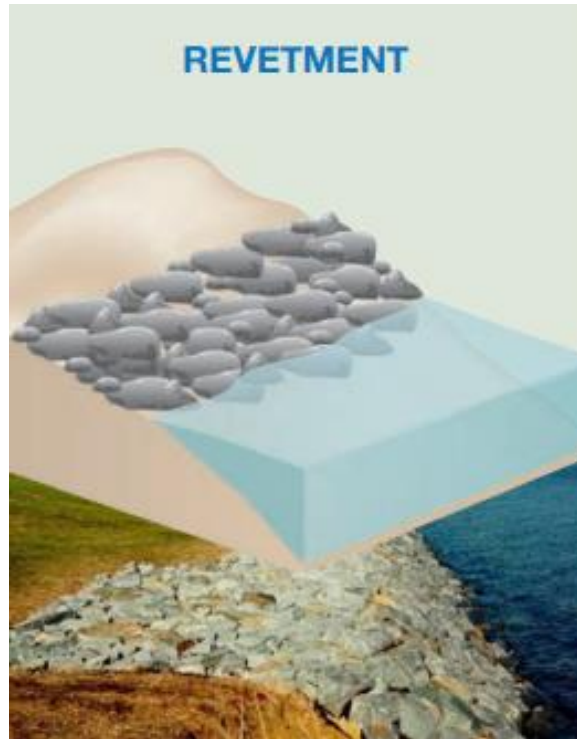
Motivation

- “Green” – Softer Techniques:



Motivation

- “Gray” – Harder Techniques:



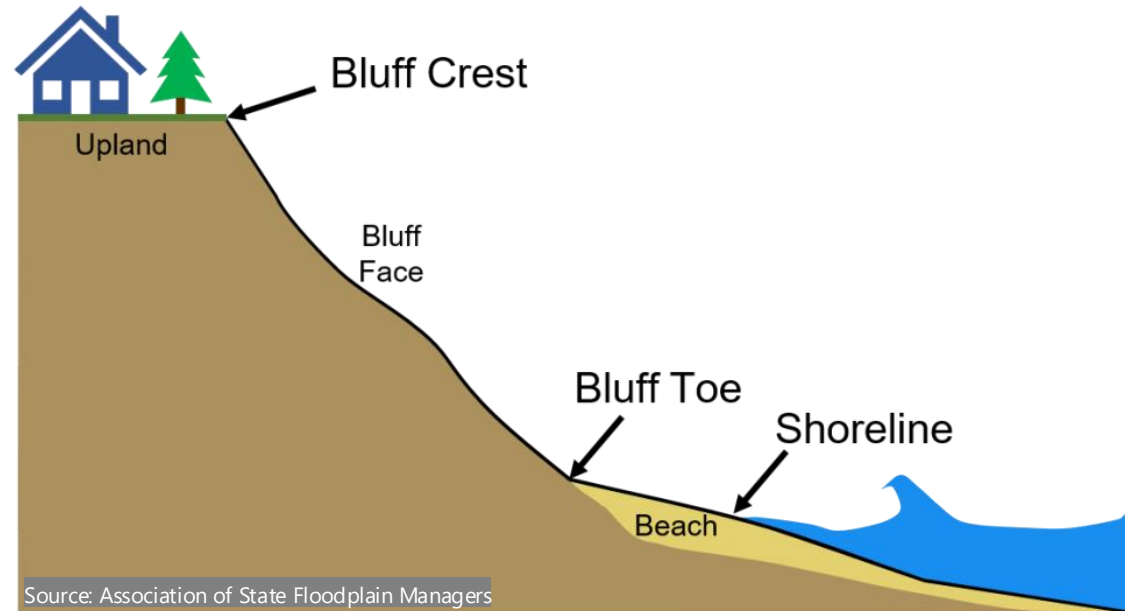
Motivation

- Bluff-protection structures can help prevent events like this:

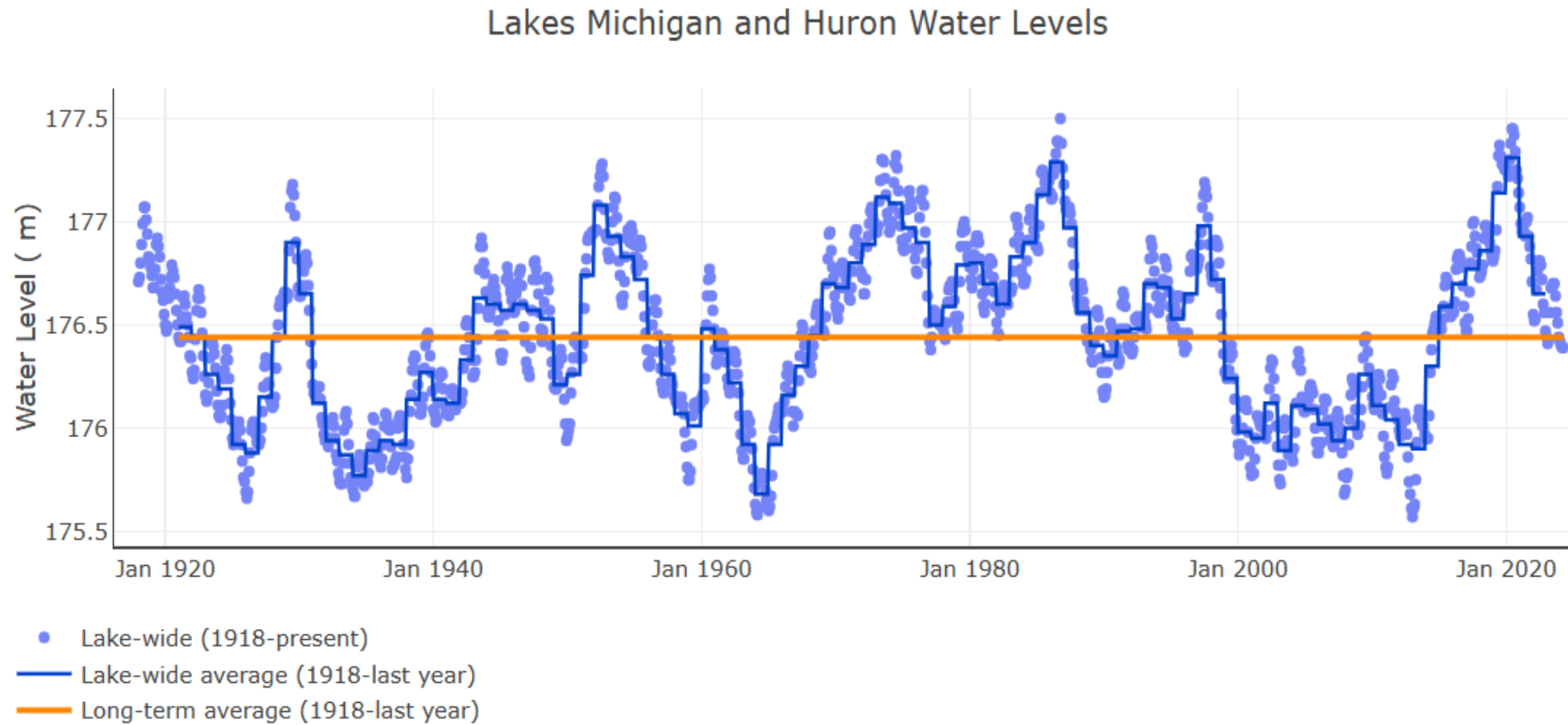


Motivation

- Rise in lake water levels brings the waves closer to the toe of the bluff, causing it to destabilize and eventually fail (Mangham, 2017, pg. 4., & Struck et al. 2021).
 - Bluff-protection structures can help mitigate the toe erosion done by the waves (NOAA, 2021, pg. 3-6).



Motivation



Outline

- Motivation: Bluff-protection structures can help mitigate bluff failure.
- Background: Spatial analysis process.
- Case Study: Ozaukee County Lake Michigan bluffs.
- Conclusions: What did we learn?
- Future work: What do we need to learn?



Background

- How did we conduct the spatial analysis?



- Gathering data:



OpenTopography
High-Resolution Topography Data and Tools



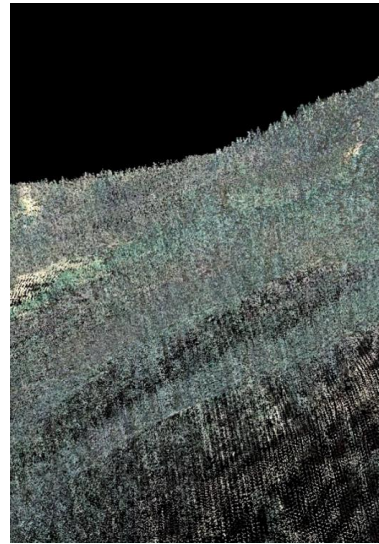
STATE CARTOGRAPHER'S OFFICE

Background

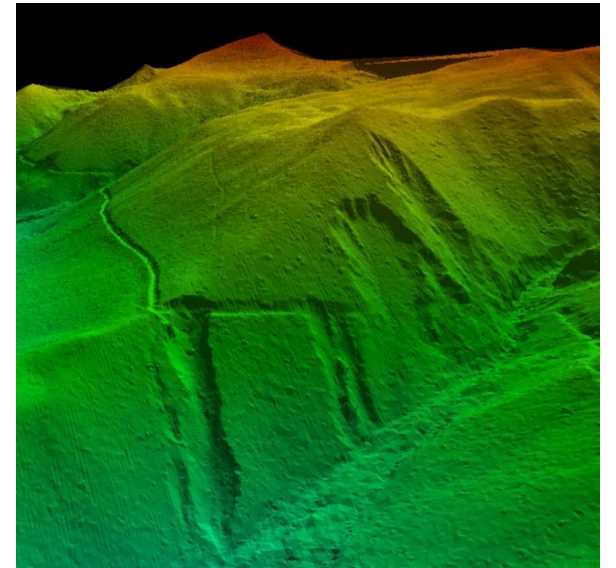
- What are Digital Elevation Models (DEMs)?



Light Detection and Ranging
(**LIDAR**)



Point Cloud Data



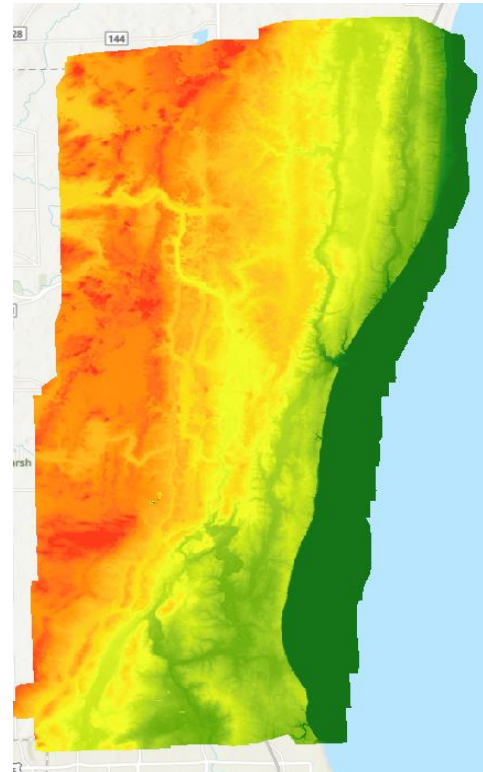
Digital Elevation Models
(**DEM**)

Background

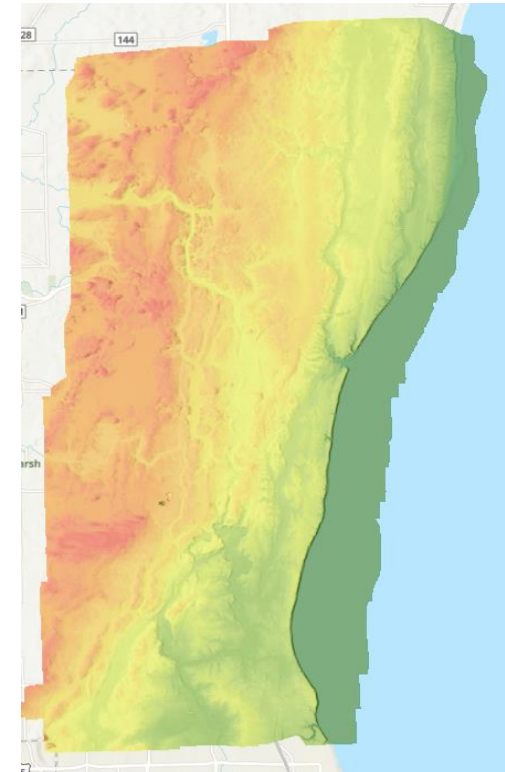
- After we gathered the DEMs, we visualized them in ArcGIS Pro:



Hillshade Only



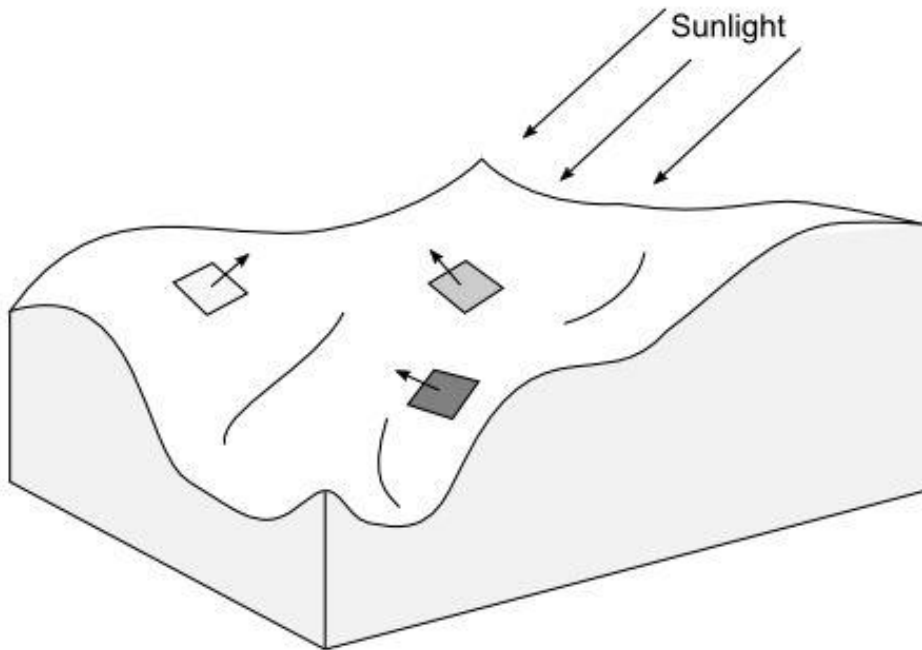
DEM Only



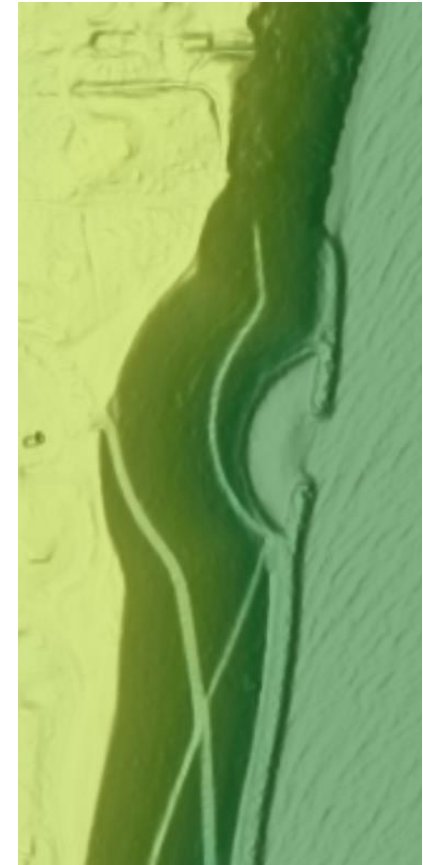
Hillshade and DEM

Background

- What is a Hillshade?
 - Makes 3D representation of the terrain surface using a light from a specified position to shade the DEM.



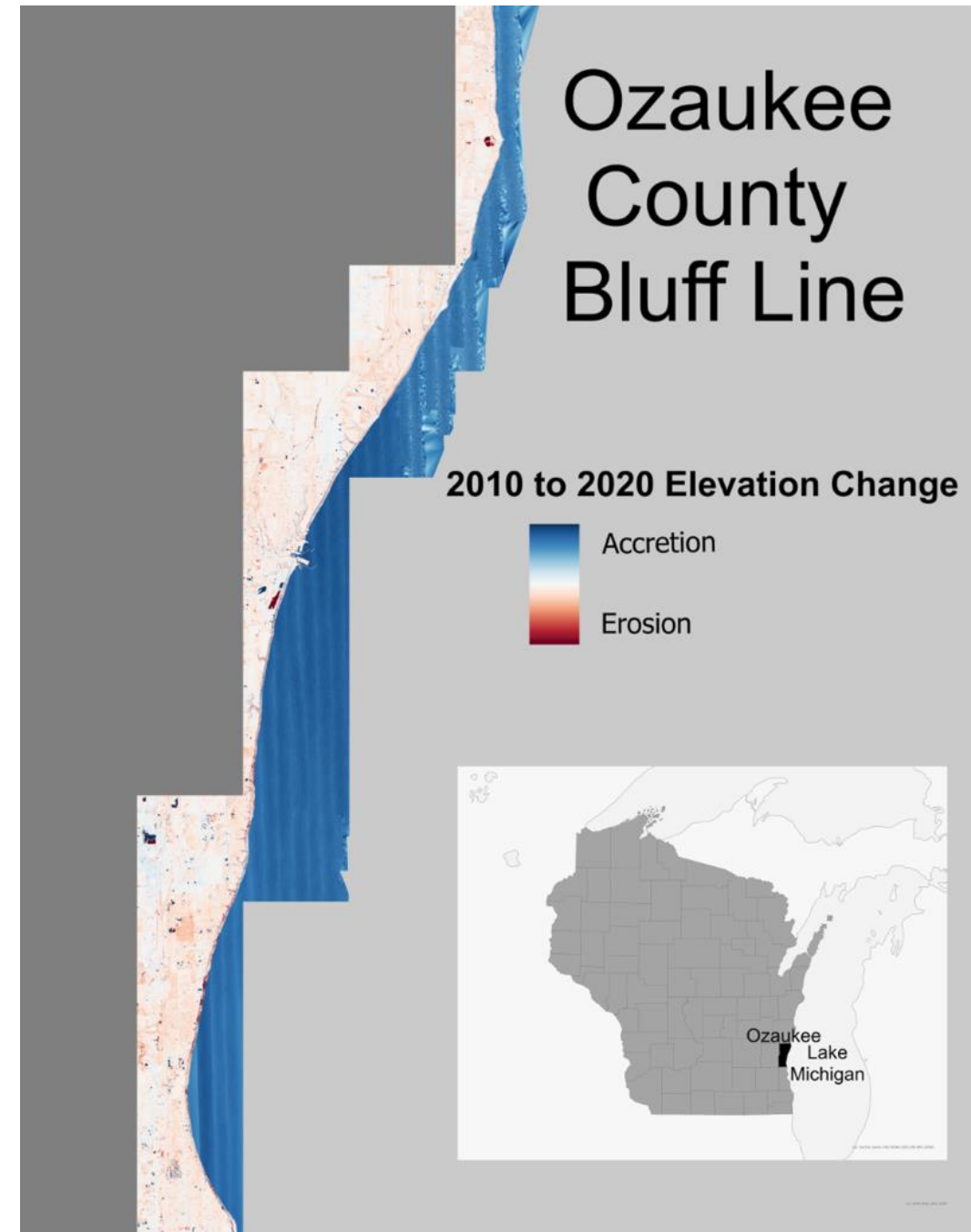
No Hillshade



Hillshade

Background

- Erosion and accretion map:
 - Raster calculator:
 - $2010 - 2020 =$ 



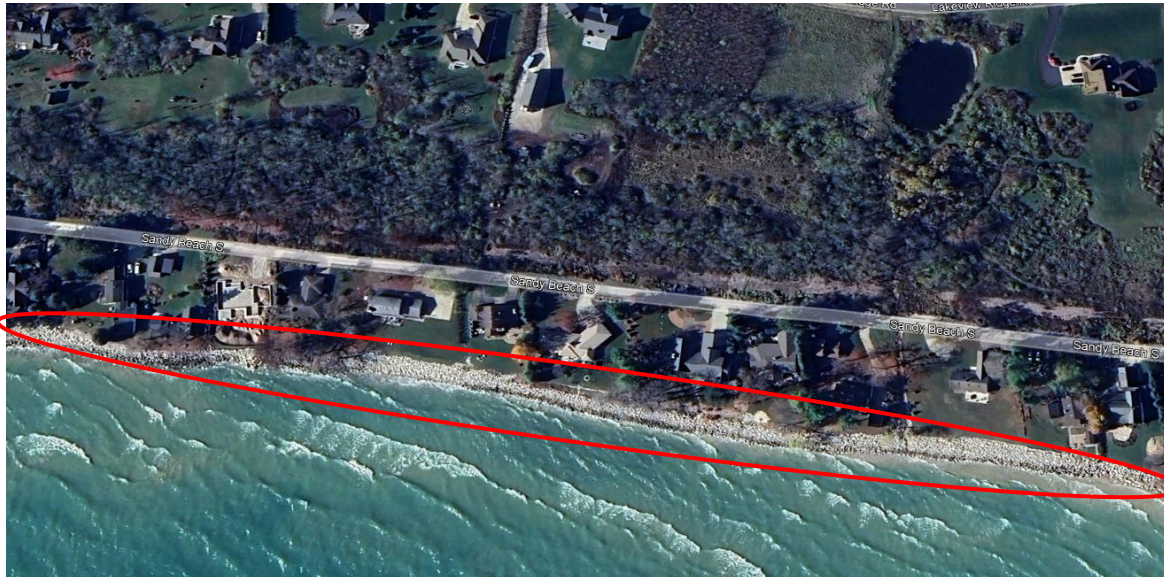
Outline

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Case study



- Google Earth Pro satellite imagery used to locate structures.

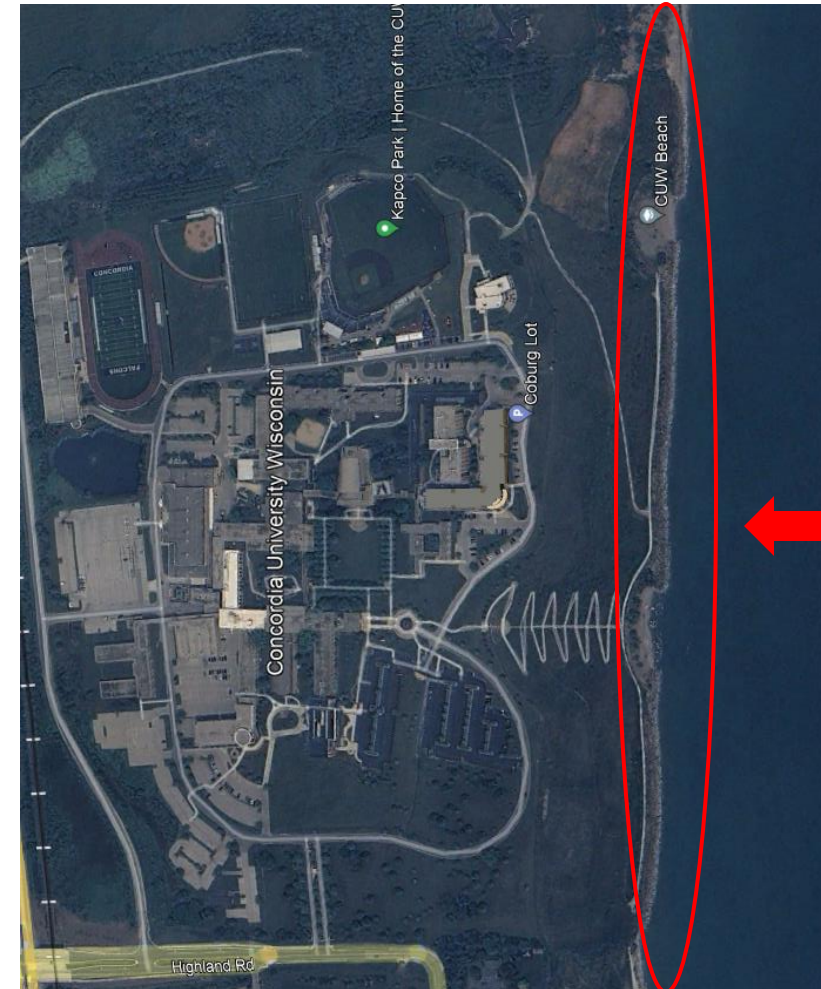


Case study



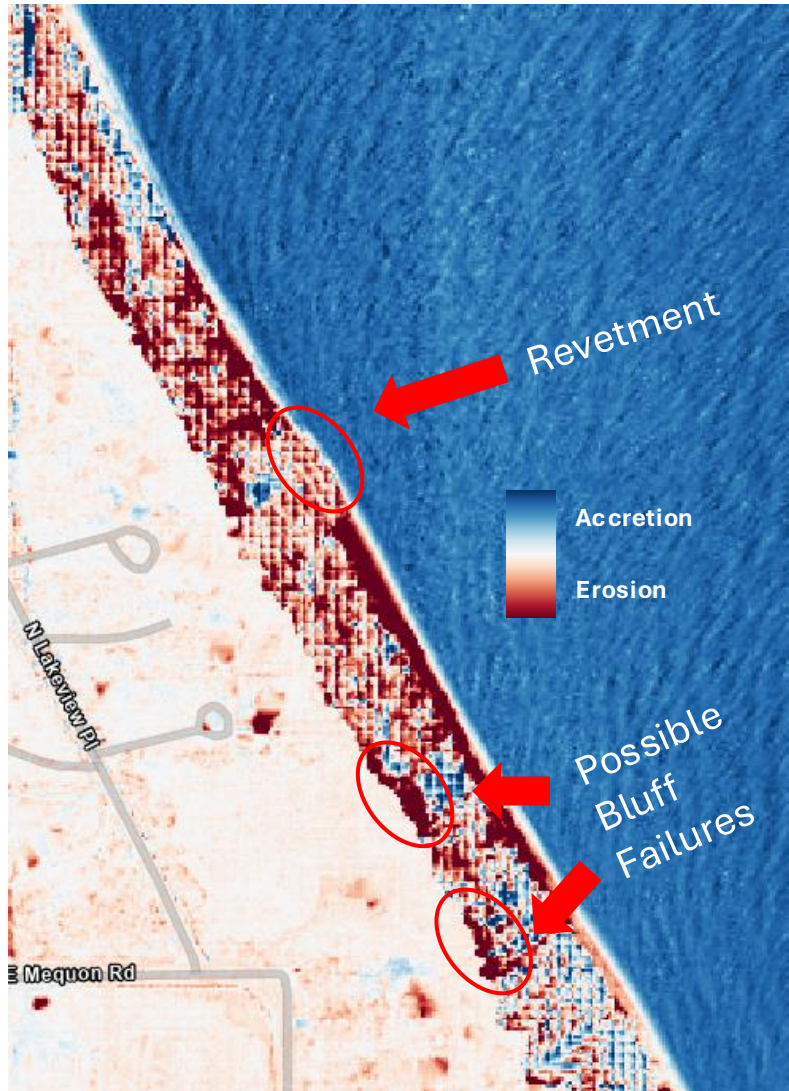
Beach
Recreational
Area Erosion.

Beach
Recreational
Area Erosion.

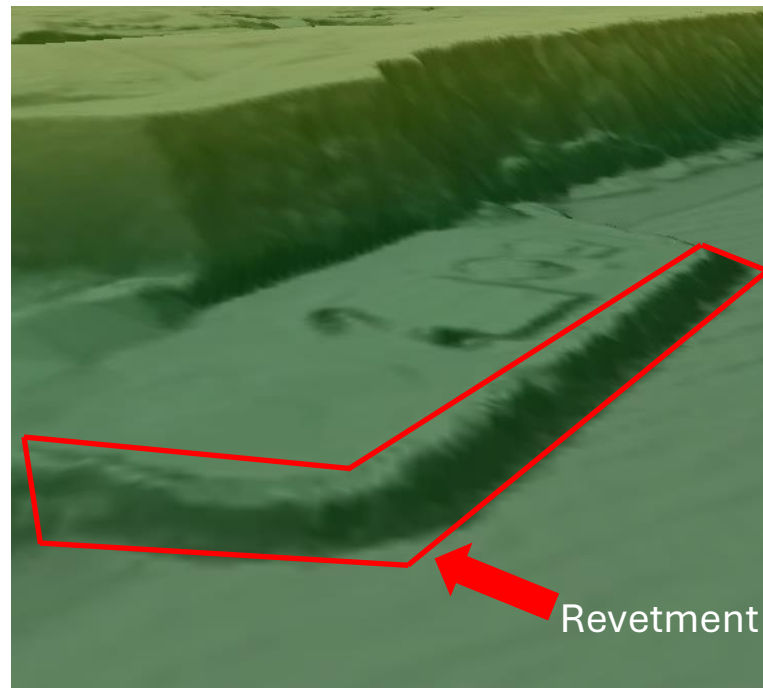
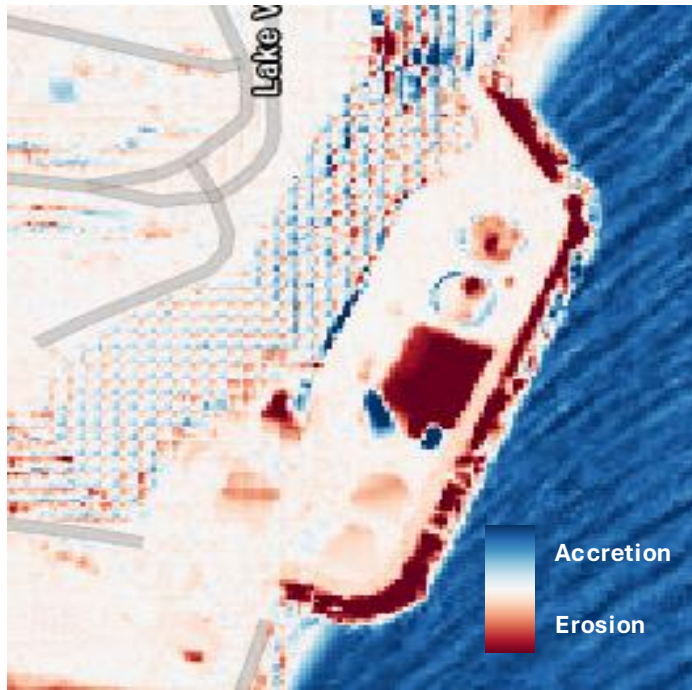


Revetment

Case study



Case study



2010



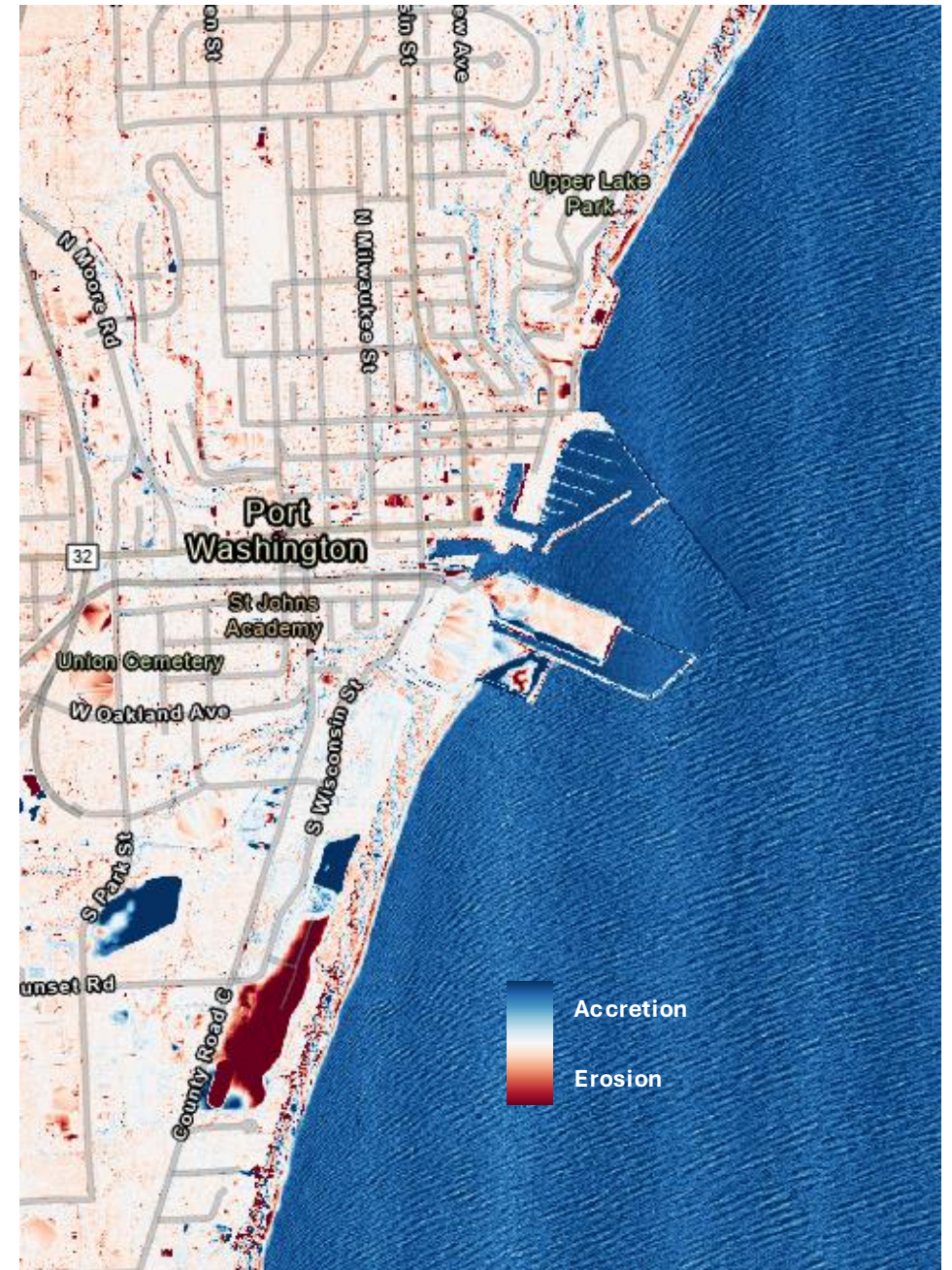
2020

Outline

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Conclusions: What did we learn?

- About the spatial analysis:
 - Can be used to aid the assessment of erosion of areas of interest.
- About the Ozaukee County Lake Michigan Bluffs:
 - The implemented revetment seems to aid the mitigation of erosion.



Outline

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Future work: What do we need to learn?

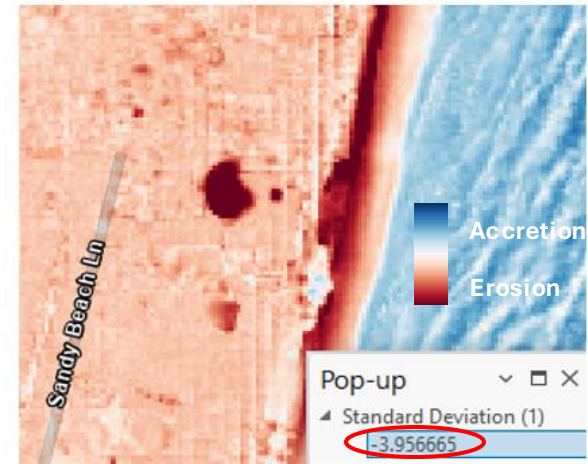
- Many factors can contribute to the erosion of the bluffs toes:
 - Lake water levels (Mangham, 2017, pg. 4., & Struck et al. 2021).
 - Intensity and frequency of storms (Lulloff et al., 2011, pg. 12).
 - The material the bluffs are composed of (Lulloff et al., 2011, pg. 12).



2010



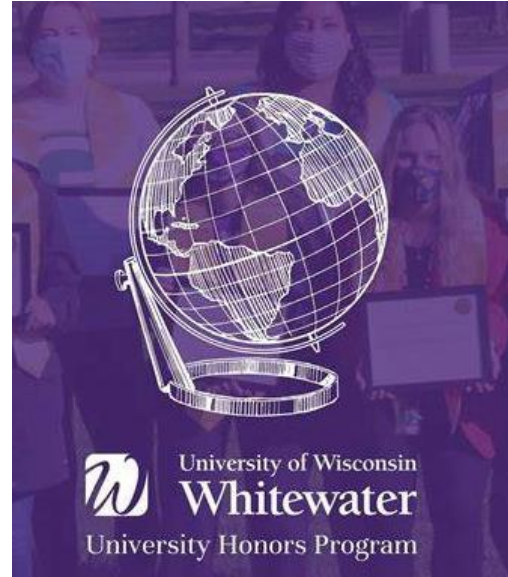
2020



Thank You!

Dr. Eric Compas

Dr. Tanya Kam



UMHC

References

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Questions?